

NETWORK TRAFFIC ANALYSIS REPORT

Wireshark Packet Capture & Protocol Analysis

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Date:
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Tool Used:
Wireshark 4.6.6
Environment:
Home Wi-Fi Network — Windows 11
Capture File
capture1.pcapng

1. Objective

The purpose of this exercise was to:

- Capture live network traffic from a personal laptop using Wireshark
- Identify and classify protocols observed in the capture
- Map each protocol to its OSI model layer and port number
- Trace a complete DNS query from request to response
- Identify a TCP 3-way handshake and understand its security significance

2. Environment & Setup

All traffic was captured on a personal home network. No third-party or enterprise networks were used.

Parameter	Details
Operating System	Windows 11
Wireshark Version	4.6.6
Network Interface Used	Wi-Fi

Network Type	Home Wi-Fi (WPA2)
Capture File Name	capture1.pcapng
Total Packets Captured	1653 packets
Capture Duration	142.2 seconds

3. Protocols Observed

The following protocols were identified using Wireshark display filters. Type each filter name in the filter bar to isolate that protocol.

Protocol	Filter Used	Port	OSI Layer	What I Saw	SOC Relevance
DNS	dns	UDP 53	Layer 7 — Application	Queries to google.com, mail.google.com	DNS tunnelling used for data exfiltration & C2
TLS/HTTPS	tls	TCP 443	Layer 7 — Application	Client Hello packets like SNI:mail.google.com	Malware hides inside encrypted TLS traffic
HTTP	http	TCP 80	Layer 7 — Application	None Observed	Credentials visible in plain text — easy to sniff
TCP	tcp.flags.syn==1	Various	Layer 4 — Transport	SYN packets to port 443	SYN floods = DoS attack; many SYNs = port scan
ICMP	icmp	None	Layer 3 — Network	Port Unreachable	ICMP tunnelling used for covert data transfer
ARP	arp	None	Layer 2 — Data Link	None Observed	ARP spoofing enables Man-in-the-Middle attacks

4. IP Address Analysis

4.1 Private vs Public IPs Observed

Every packet has a Source and Destination IP address. Private IPs belong to your home network. Public IPs belong to servers on the internet.

IP Address	Type	Belongs To
192.168.0.175	Private	My laptop
192.168.0.1	Private	My home router / Wi-Fi gateway
142.250.71.106	Public	Google LLC
151.101.65.91	Public	Fastly Inc.
52.110.16.165	Public	Microsoft Corporation

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	142.250.206.106	192.168.0.175	UDP	278	443 → 65181 Len=236
2	0.000778400	142.250.206.106	192.168.0.175	UDP	278	443 → 65181 Len=236
3	0.001196100	192.168.0.175	142.250.206.106	UDP	80	65181 → 443 Len=38
4	1.449674400	192.168.0.175	142.250.71.106	TCP	55	57590 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
5	1.461765000	142.250.71.106	192.168.0.175	TCP	66	443 → 57590 [ACK] Seq=1 Ack=2 Win=384 Len=0 SLE=1 SRE=2
6	2.224259700	192.168.0.175	192.168.0.1	DNS	77	Standard query 0xf88c HTTPS cdn.growthbook.io
7	2.226927500	192.168.0.175	192.168.0.1	DNS	77	Standard query 0x8246 A cdn.growthbook.io
8	2.227280100	192.168.0.175	151.101.65.91	TLSv1.2	128	Application Data
9	2.227378900	192.168.0.175	151.101.65.91	TLSv1.2	93	Application Data
10	2.235870800	151.101.65.91	192.168.0.175	TCP	54	443 → 56683 [ACK] Seq=1 Ack=114 Win=192 Len=0
11	2.253654000	151.101.65.91	192.168.0.175	TLSv1.2	93	Application Data
12	2.253655900	151.101.65.91	192.168.0.175	TLSv1.2	152	Application Data
13	2.253656700	151.101.65.91	192.168.0.175	TLSv1.2	107	Application Data
14	2.253815700	192.168.0.175	151.101.65.91	TCP	54	56683 → 443 [ACK] Seq=114 Ack=191 Win=252 Len=0
15	2.255667300	192.168.0.175	151.101.65.91	TLSv1.2	89	Application Data
16	2.270187500	192.168.0.1	192.168.0.175	DNS	338	Standard query response 0xf88c HTTPS cdn.growthbook.io CNAME n.sni.global.fastly.net NS l.root-servers.net NS m.root-s...
17	2.272927900	192.168.0.1	192.168.0.175	DNS	402	Standard query response 0x8246 A cdn.growthbook.io CNAME n.sni.global.fastly.net A 151.101.129.91 A 151.101.1.91 A 151...
18	2.282462900	151.101.65.91	192.168.0.175	TCP	54	443 → 66683 [ACK] Seq=91 Ack=191 Win=192 Len=0
19	3.270319800	192.168.0.175	192.168.0.1	DNS	92	Standard query 0xc200 A mobile.events.data.microsoft.com
20	3.288152400	192.168.0.1	192.168.0.175	DNS	437	Standard query response 0xc200 A mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME on...
21	3.292077400	192.168.0.175	51.132.193.104	TCP	66	63737 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
22	3.454453200	51.132.193.104	192.168.0.175	TCP	66	443 → 63737 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
23	3.454544300	192.168.0.175	51.132.193.104	TCP	54	63737 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
24	3.459729300	192.168.0.175	51.132.193.104	TLSv1.3	518	Client Hello (SN=mobile.events.data.microsoft.com)
25	3.469771800	51.132.193.104	192.168.0.175	TCP	54	443 → 63737 [ACK] Seq=1 Ack=465 Win=98048 Len=0

5. DNS Query Analysis

DNS (Domain Name System) is the internet's phonebook — it converts website names like google.com into IP addresses. Every website you visit triggers a DNS lookup. DNS runs on UDP Port 53 at OSI Layer 7.

5.1 DNS Lookup Traced End-to-End

Filter used in Wireshark: dns

Field	Query Packet (Request)	Response Packet (Reply)
Packet Number	50	51
Source IP	192.168.0.175	192.168.0.1
Destination IP	192.168.0.1	192.168.0.175
Protocol & Port	DNS — UDP Port 53	DNS — UDP Port 53
Info Column Text	Standard query A self.events.data.microsoft.com	Standard query response A self.events.data.microsoft.com

Domain Queried	self.events.data.microsoft.com	—
IP Address Returned	—	20.42.73.27

5.2 Plain English Explanation

My laptop asked the DNS server: "What is the IP address of self.events.data.microsoft.com?" The DNS server replied: "The IP address is 20.42.73.27." My browser then connected to that IP address to load the website.

5.3 Why SOC Analysts Monitor DNS

- Malware phones home using DNS queries to attacker-controlled domains
- DNS tunnelling hides stolen data inside DNS query text
- Unusually large DNS responses can indicate data being exfiltrated
- High-frequency queries to one strange domain = likely C2 beacons

No.	Time	Source	Destination	Protocol	Length	Info
6	2.224250700	192.168.0.175	192.168.0.1	DNS	77	Standard query 0xf88c HTTPS cdn.growthbook.io
7	2.226927500	192.168.0.175	192.168.0.1	DNS	77	Standard query 0x8246 A cdn.growthbook.io
16	2.270107500	192.168.0.1	192.168.0.175	DNS	338	Standard query response 0xf88c HTTPS cdn.growthbook.io CNAME n.sni.global.fastly.net NS l.root-servers.net NS m.root-s...
17	2.272927900	192.168.0.1	192.168.0.175	DNS	402	Standard query response 0x8246 A cdn.growthbook.io CNAME n.sni.global.fastly.net A 151.101.129.91 A 151.101.1.91 A 151...
19	3.270319800	192.168.0.175	192.168.0.1	DNS	92	Standard query 0xc200 A mobile.events.data.microsoft.com
20	3.288152400	192.168.0.1	192.168.0.175	DNS	437	Standard query response 0xc200 A mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME on...
50	4.288215200	192.168.0.175	192.168.0.1	DNS	90	Standard query 0xe252 A self.events.data.microsoft.com
51	4.349700800	192.168.0.1	192.168.0.175	DNS	432	Standard query response 0xe252 A self.events.data.microsoft.com CNAME self-events-data.trafficmanager.net CNAME onedsc...
85	6.637654500	192.168.0.175	192.168.0.1	DNS	77	Standard query 0xeba4 HTTPS cdn.growthbook.io
86	6.638109200	192.168.0.175	192.168.0.1	DNS	77	Standard query 0x8b4d A cdn.growthbook.io
95	6.684812200	192.168.0.1	192.168.0.175	DNS	402	Standard query response 0x8b4d A cdn.growthbook.io CNAME n.sni.global.fastly.net A 151.101.1.91 A 151.101.129.91 A 151...
96	6.694443600	192.168.0.1	192.168.0.175	DNS	402	Standard query response 0xeba4 HTTPS cdn.growthbook.io CNAME n.sni.global.fastly.net NS c.root-servers.net NS d.root-s...
154	16.927383200	192.168.0.175	192.168.0.1	DNS	77	Standard query 0x893b HTTPS gemini.google.com
155	16.927864200	192.168.0.175	192.168.0.1	DNS	77	Standard query 0x9e89 A gemini.google.com
156	16.976131400	192.168.0.1	192.168.0.175	DNS	149	Standard query response 0x893b HTTPS gemini.google.com NS ns1.google.com NS ns3.google.com NS ns4.google.com NS ns2.go...
157	16.981987700	192.168.0.1	192.168.0.175	DNS	277	Standard query response 0x9e89 A gemini.google.com A 142.251.152.2 A 142.251.153.2 A 142.251.155.2 A 142.251.156.2 A 1...
245	29.939071900	192.168.0.175	192.168.0.1	DNS	86	Standard query 0xcd79 HTTPS waa-pa.clients6.google.com
246	29.939466700	192.168.0.175	192.168.0.1	DNS	86	Standard query 0xa28d A waa-pa.clients6.google.com
251	30.010443300	192.168.0.1	192.168.0.175	DNS	158	Standard query response 0xcd79 HTTPS waa-pa.clients6.google.com NS ns4.google.com NS ns2.google.com NS ns1.google.com ...
253	30.024064200	192.168.0.1	192.168.0.175	DNS	174	Standard query response 0xa28d A waa-pa.clients6.google.com A 192.178.174.95 NS ns2.google.com NS ns1.google.com NS ns...
377	45.921491700	192.168.0.175	192.168.0.1	DNS	79	Standard query 0x6507 HTTPS accounts.google.com
378	45.924105000	192.168.0.175	192.168.0.1	DNS	79	Standard query 0x156a A accounts.google.com
379	45.931632400	192.168.0.1	192.168.0.175	DNS	167	Standard query response 0x156a A accounts.google.com A 192.178.211.84 NS ns2.google.com NS ns1.google.com NS ns3.googl...
383	45.955709500	192.168.0.1	192.168.0.175	DNS	151	Standard query response 0x6507 HTTPS accounts.google.com NS ns4.google.com NS ns2.google.com NS ns1.google.com NS ns3...
408	46.126807300	192.168.0.175	192.168.0.1	DNS	80	Standard query 0x04f1 HTTPS beacons.gcp.gvt2.com

6. TCP 3-Way Handshake

Before any data is exchanged over TCP, two devices must complete a 3-step handshake to establish a reliable connection. The steps are: SYN → SYN-ACK → ACK.

6.1 Handshake Traced

Filter used in Wireshark: tcp.flags.syn==1

Step	Packet #	Source IP	Destination IP	Dest. Port	TCP Flags	Meaning
1 — SYN	52	192.168.0.175	20.42.73.27	443	SYN	My laptop says: I want to connect
2 — SYN-ACK	58	20.42.73.27	192.168.0.175	63738	SYN + ACK	Server says: OK, connection accepted
3 — ACK	59	192.168.0.175	20.42.73.27	443	ACK	My laptop says: Confirmed, let us begin

6.2 Why SOC Analysts Monitor TCP Handshakes

- Many SYN packets with NO SYN-ACK replies = Port scan (attacker mapping the network)
- Flood of SYN packets from one IP = SYN Flood DDoS attack
- Connection to unusual port like 4444 or 1337 = possible malware reverse shell
- Half-open connections that never complete = sign of reconnaissance activity

No.	Time	Source	Destination	Protocol	Length	Info
52	4.353140000	192.168.0.175	20.42.73.27	TCP	66	63738 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
58	4.590955800	20.42.73.27	192.168.0.175	TCP	66	443 → 63738 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
59	4.591145100	192.168.0.175	20.42.73.27	TCP	54	63738 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
60	4.598027100	192.168.0.175	20.42.73.27	TLSv1.3	516	Client Hello (SNL=self.events.data.microsoft.com)
61	4.607675900	20.42.73.27	192.168.0.175	TCP	54	443 → 63738 [ACK] Seq=1 Ack=463 Win=98048 Len=0
62	4.607680500	20.42.73.27	192.168.0.175	TCP	54	[TCP Dup ACK 61#1] 443 → 63738 [ACK] Seq=1 Ack=463 Win=98048 Len=0
63	4.825161900	20.42.73.27	192.168.0.175	TLSv1.3	1514	Server Hello, Change Cipher Spec
64	4.825165600	20.42.73.27	192.168.0.175	TCP	1514	443 → 63738 [PSH, ACK] Seq=1461 Ack=463 Win=98304 Len=1460 [TCP PDU reassembled in 69]
65	4.825167300	20.42.73.27	192.168.0.175	TCP	1514	443 → 63738 [PSH, ACK] Seq=2921 Ack=463 Win=98304 Len=1460 [TCP PDU reassembled in 69]
66	4.825168600	20.42.73.27	192.168.0.175	TCP	1514	443 → 63738 [PSH, ACK] Seq=4381 Ack=463 Win=98304 Len=1460 [TCP PDU reassembled in 69]
67	4.825170700	20.42.73.27	192.168.0.175	TCP	1514	443 → 63738 [PSH, ACK] Seq=5841 Ack=463 Win=98304 Len=1460 [TCP PDU reassembled in 69]
68	4.825172000	20.42.73.27	192.168.0.175	TCP	1514	[TCP Retransmission] 443 → 63738 [PSH, ACK] Seq=5841 Ack=463 Win=98304 Len=1460 [TCP PDU reassembled in 69]
69	4.825188200	20.42.73.27	192.168.0.175	TLSv1.3	1260	Application Data
70	4.825332600	192.168.0.175	20.42.73.27	TCP	66	63738 → 443 [ACK] Seq=463 Ack=7301 Win=262144 Len=0 SLE=5841 SRE=7301
71	4.825416300	192.168.0.175	20.42.73.27	TCP	54	63738 → 443 [ACK] Seq=463 Ack=8507 Win=260864 Len=0
72	4.848331300	192.168.0.175	20.42.73.27	TLSv1.3	134	Change Cipher Spec, Application Data
73	4.848794800	192.168.0.175	20.42.73.27	TLSv1.3	580	Application Data
74	4.848850600	192.168.0.175	20.42.73.27	TCP	1494	63738 → 443 [ACK] Seq=1069 Ack=8507 Win=260864 Len=1440 [TCP PDU reassembled in 75]
75	4.848850600	192.168.0.175	20.42.73.27	TLSv1.3	605	Application Data
76	4.869396600	20.42.73.27	192.168.0.175	TCP	54	443 → 63738 [ACK] Seq=8507 Ack=1069 Win=97792 Len=0
77	4.869397800	20.42.73.27	192.168.0.175	TCP	54	443 → 63738 [ACK] Seq=8507 Ack=3060 Win=96000 Len=0
78	4.869398100	20.42.73.27	192.168.0.175	TLSv1.3	157	Application Data
79	5.086402700	192.168.0.175	20.42.73.27	TCP	54	63738 → 443 [ACK] Seq=3060 Ack=8610 Win=260608 Len=0
80	5.086482000	20.42.73.27	192.168.0.175	TLSv1.3	500	Application Data
81	5.086800000	192.168.0.175	20.42.73.27	TCP	54	63738 → 443 [ACK] Seq=3060 Ack=9056 Win=262144 Len=0

7. Key Observations & Findings

The following observations were noted during the analysis. These are my personal findings from the actual capture.

#	What I Observed	Protocol	Normal or Suspicious?	My Reasoning
1	DNS query for self.events.data.microsoft.com returned IP 20.42.73.27	DNS	Normal	Known Microsoft Azure server — Windows sends telemetry data automatically

2	Encrypted HTTPS traffic to multiple IPs	TLS	Normal	Standard HTTPS — modern browsers always use encrypted connections
3	No ARP packets observed during capture duration	ARP	Normal	ARP broadcasts are infrequent — short capture may not include them
4	Laptop sent DNS queries in background without me actively browsing	DNS	Normal	Background Windows services and apps make automatic DNS requests — common OS behaviour
5	Multiple TLS Client Hello packets sent to different IPs within seconds	TLS	Normal	Each website loads from multiple servers simultaneously — each connection starts with a Client Hello on port 443

8. Real World Application

Through this exercise I understood how these network concepts apply to real cybersecurity work:

Skill Practised in This Lab	How It Applies in a Real SOC Role
Reading and filtering packet captures	SOC analysts investigate .pcap files attached to security incidents and alerts
Identifying protocols by port number	SIEM alerts display port numbers — knowing what runs on each port speeds triage
Tracing DNS queries	Malware uses DNS for command-and-control — analysts flag unusual domain lookups
Recognising TCP handshake patterns	Port scans and SYN floods appear in SIEM — recognising patterns is essential for Tier 1
Distinguishing private vs public IPs	Identifies whether an attack is coming from inside or outside the network
Writing structured analysis reports	SOC analysts document every investigation — clear reports support escalation and audits

9. Tools Used

Tool	Version	Purpose	Cost
Wireshark	4.6.6	Live packet capture and analysis	Free / Open Source

whatismyipaddress.com	Online	IP address lookup and owner identification	Free
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10. Lessons Learned

Before this exercise, I opened Wireshark for the first time and could not understand anything I saw as hundreds of packets were flying across the screen with no clear meaning. The volume of data and technical terms were completely overwhelming at first. After learning the concepts behind networking such as the OSI model, protocols, port numbers and IP addressing, I revisited Wireshark with a better approach. Using display filters like `dns`, `tls`, `arp` and `tcp.flags.syn==1`, I was able to isolate specific traffic and understand what each packet was doing. I learned that raw packet data is not meant to be read all at once and that filters are the key to finding what matters. I now understand how a DNS query works end to end, how a TCP 3-way handshake establishes a connection and how to distinguish normal background traffic from suspicious activity. This project gave me a strong practical foundation for my journey toward becoming a SOC Analyst.